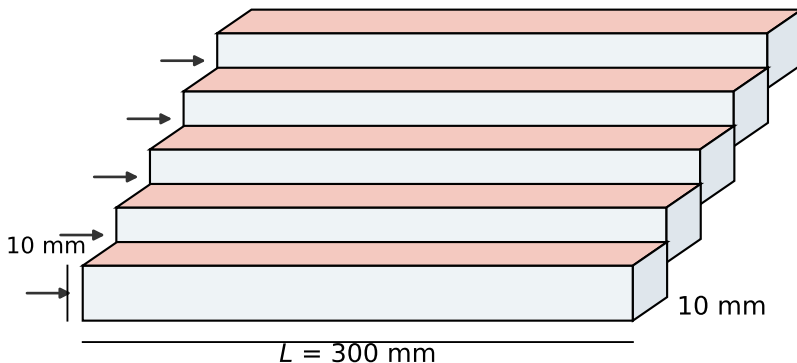


(a) cold-plate heat-exchanger domain

heated wall (top of every duct): uniform $q_w = \dot{Q}/A_{heated} = 627 \text{ W m}^{-2}$

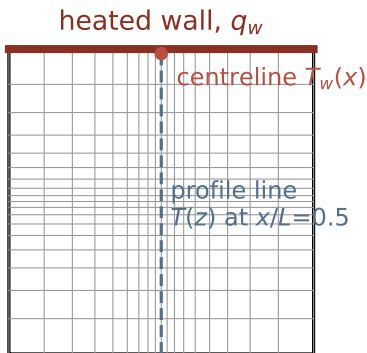


inlet: T_{in} , $\dot{m}$ (He: mass-flow BC; LN₂: velocity BC); outlet: fixed p

other walls adiabatic and no slip; five square ducts, $D_h = 10 \text{ mm}$, pitch 18 mm

heated area 0.015 m^2 (top walls), flow area $5 \times 10^{-4} \text{ m}^2$

(b) duct cross-section: mesh and sampling lines



18×18 cells per duct

two-sided grading, ratio 6

first cell 0.19 mm

180 cells along L

2.9×10^5 cells in total